

SANA NIROOMAND

Computer Engineering Student | AI Engineering, Biomedical Informatics

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EDUCATION

B.S. in Computer Engineering

Sharif University of Technology

📅 Sept 2023 – Present

- **GPA:** 18.94/20.0 (3.79/4.0)
- **Coursework:** ML, AI, Biomedical Image Analysis, Computational Drug Design, Bioinformatics, Algorithms, Data Structures, Linear Algebra, Probability.

WORK

Research Intern – HDR Object Detection

Middle East Technical University (METU)

📅 Jul 2026 – August 2026

📍 Ankara, Turkey

- Conducting research under the supervision of Prof. Aydin Alatan to determine whether object detection performs better directly on HDR data or after HDR-to-LDR tone mapping.
- Designing and running comparative experiments on the local HDR4RTT dataset to evaluate how tone-mapping stages affect object-detection performance.

AI EXPERIENCE

PI-CAI Prostate Cancer Prediction

Medical Imaging / Weak Supervision / MIL

- Built an interpretable, uncertainty-aware MIL pipeline for prostate cancer prediction using PI-CAI-style MRI data.
- Adapted multi-scale attention-based aggregation for weak supervision where lesion-level annotations are limited.
- Focused on interpretability, uncertainty estimation, and evaluation for medical imaging workflows.

DD-PRiSM-plus – Synergistic Drug Combination Prediction

Deep Learning / Drug Discovery / Reproducible ML

- Co-developed a reproducible, leakage-safe DD-PRiSM pipeline on NCI60 and NCI-ALMANAC for predicting cancer monotherapy and drug-combination responses, matching published results within 0.0016 RMSE.
- Fused ChemBERTa embeddings with Morgan fingerprints to raise unseen-drug PCC from 0.7585 to 0.7912 (6.4% lower RMSE); vectorized the PyTorch model for 5.2× faster training on a Tesla T4.

Transcriptomic Analysis of Psoriasis and Psoriatic Arthritis Skin

Bioinformatics / RNA-seq

- Reproduced parts of a transcriptomics study using public RNA-seq datasets.
- Worked on differential gene expression analysis and interpretation of immune-related transcriptional patterns.

Chorus – AI-Powered Language Learning Bot

Python / Generative AI / NLP

- Developed a multilingual Telegram bot that transforms songs into adaptive language-learning exercises using the OpenAI and LRCLIB APIs.
- Implemented lyric retrieval, audio metadata extraction, fill-in-the-blank generation, answer evaluation with explanations, beginner word banks, and cultural-context guidance.

ACHIEVEMENTS

- **Top 0.14% Nationally**
Iranian Nationwide University Entrance Exam in Mathematics, among 147,317 participants.
- **Gold Medal**
International exhibition in South Korea for a botanical antimicrobial spray project.
- **2nd Place**
Inventions & Achievements Track at the Avicenna Festival.
- **Provincial Distinguished Rank**
Khwarizmi Youth Festival.

SKILLS

Python

C++

C

Java

PyTorch

NumPy

Pandas

Scikit-learn

Computer Vision

Deep Learning

Biomedical AI

Bioinformatics

Drug Design

Git

Linux

LANGUAGES

- **English:** Fluent - IELTS 8
- **French:** Conversational